

Final Exam Tiny House Project

John Krogman and Conner Glover

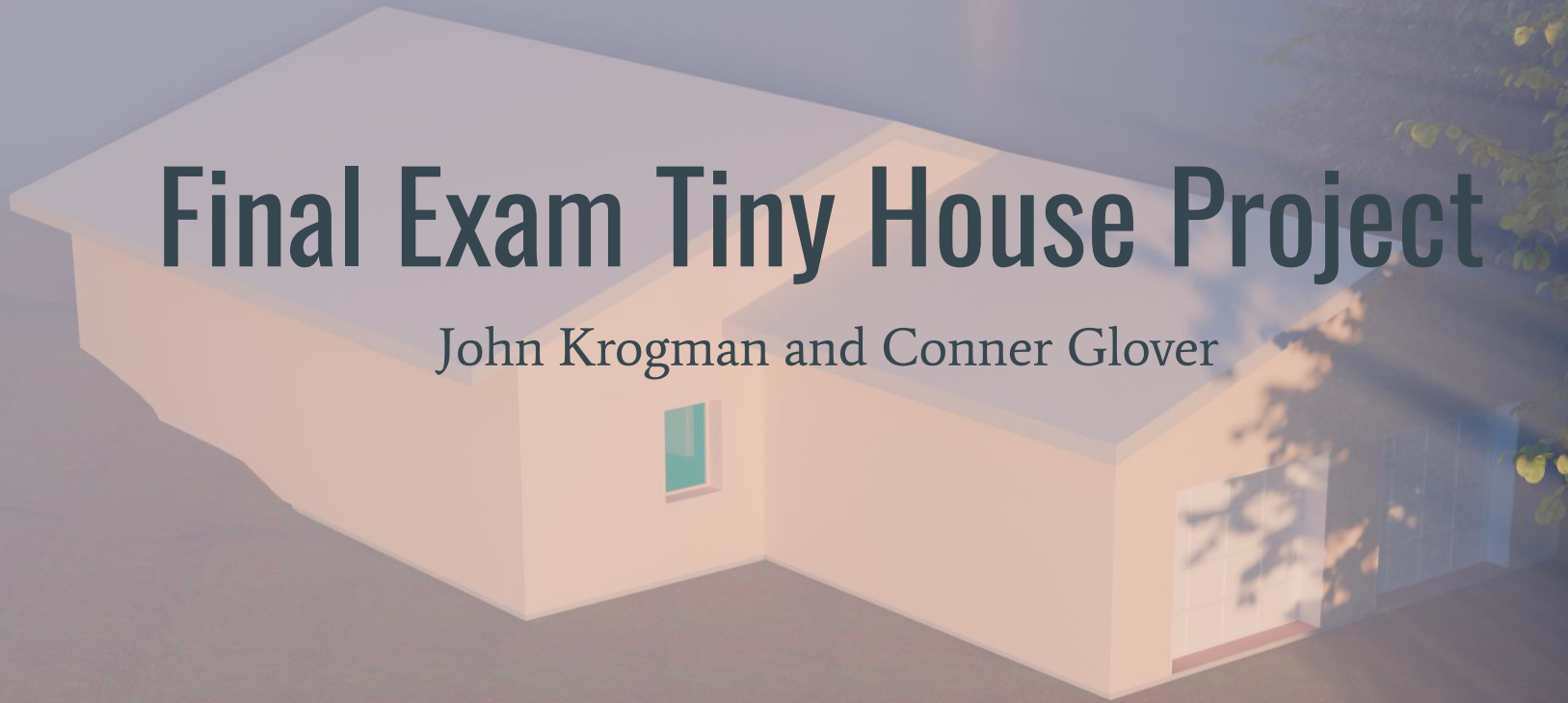


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Project Description

We were tasked with building a Habitat for Humanity home for our client. Some topics we had to consider were affordability, accessibility, energy and water efficiency, and safety.



Client

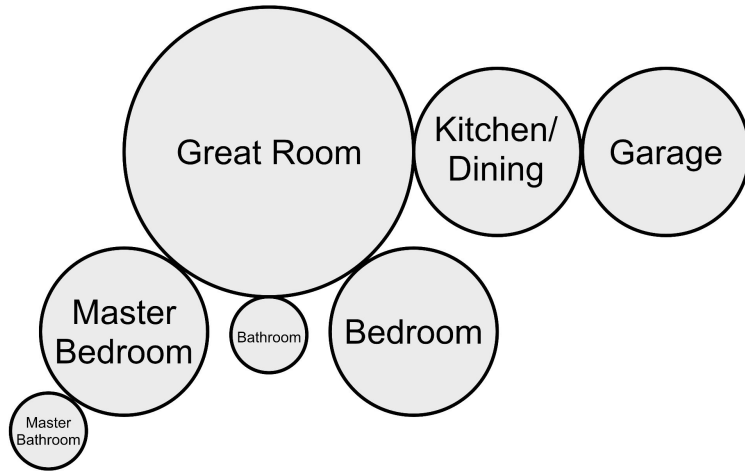
David and Sophia are a couple seeking a stable, affordable home for their eight-year-old son, Matthew. They envision a simple, energy-efficient, ranch-style house that features a garden for Sophia and space for Matthew to play soccer. This Habitat for Humanity home would provide the safety and permanent foundation their family needs.

PLTW|Engineering

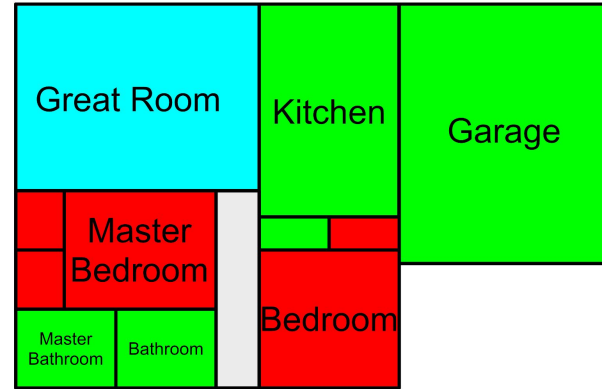
Client Survey

Family Information	
Adult Names/Ages	David (34) and Sophia (32)
Occupations	Delivery Driver (David), Preschool Teacher (Sophia)
Child Names/Ages	Matthew (8)
Physical Disabilities	
Other Special Needs	Matthew has asthma
Pets	Mittens (cat)
Architectural Details	
House Style	Simple ranch
Number of Bedrooms	2 Bedrooms (Parents, Matthew)
Number of Bathrooms	2 Bathrooms
Square Footage	1,500 - 2,000 sq. ft.
Deck or Patio	Small deck
Extra Storage	Garage or Shed
Leisure Activities	
Hobbies	Gardening (Sophia), Soccer/Legos (Matthew)
Entertainment	Family movie night
Equipment	Basic gardening tools, Kid's bicycles, soccer balls
Special Needs	
Disabilities/Illness	Asthma: Easy to clean
Energy Saving/ LEED Concepts and Ideas	
Site Development	Position to use natural light. High quality appliances. Double flush toilets. Low-water shower heads
Water Savings	
Energy Efficiency	
Materials Selection	
Indoor Environmental Quality	
Other Ideas	

Sketches



BLUE: Living
RED: Sleeping
GREEN: Utility



Floor Planner

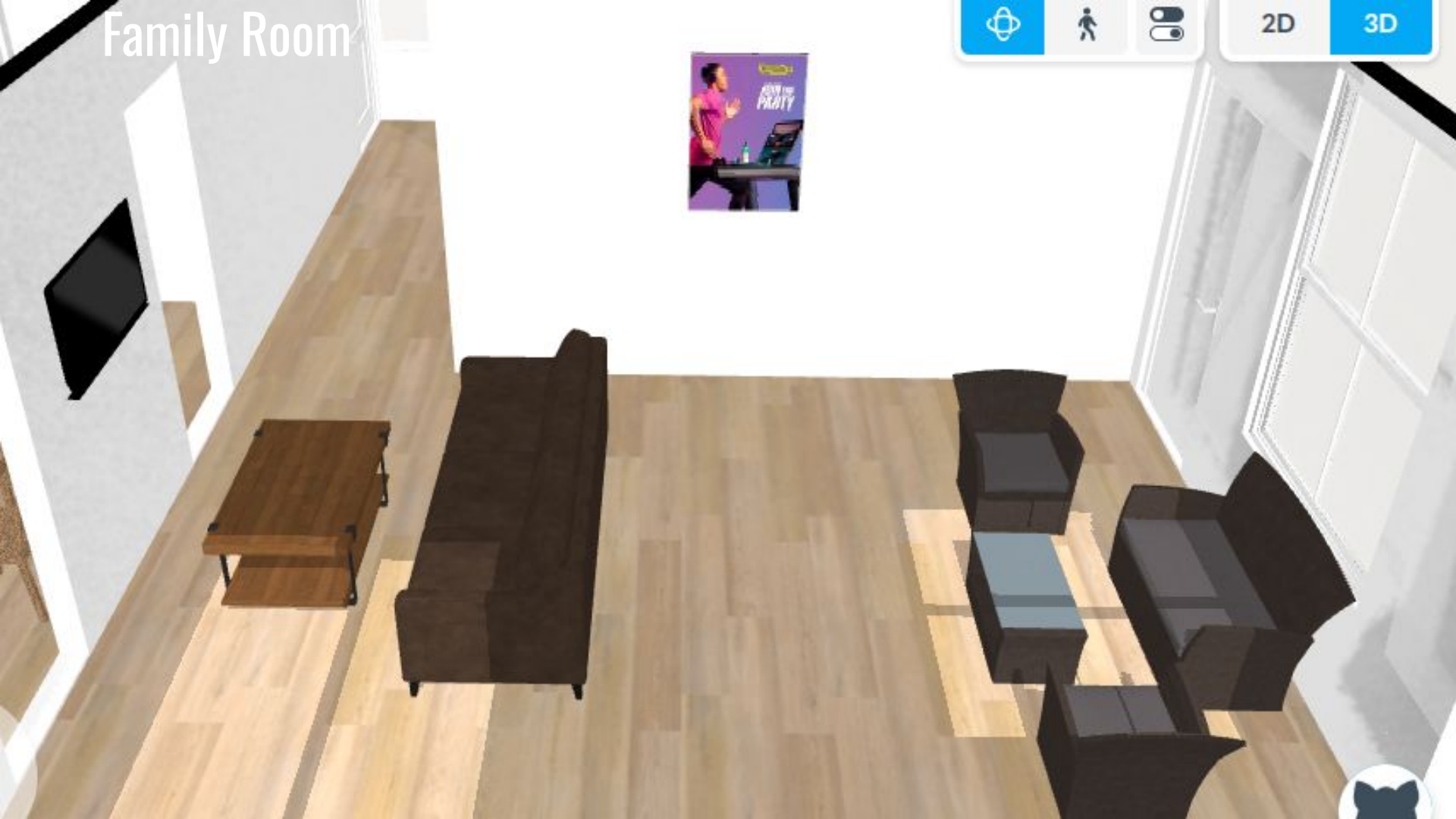
Top View



Great Room



Family Room



Kitchen



Garage



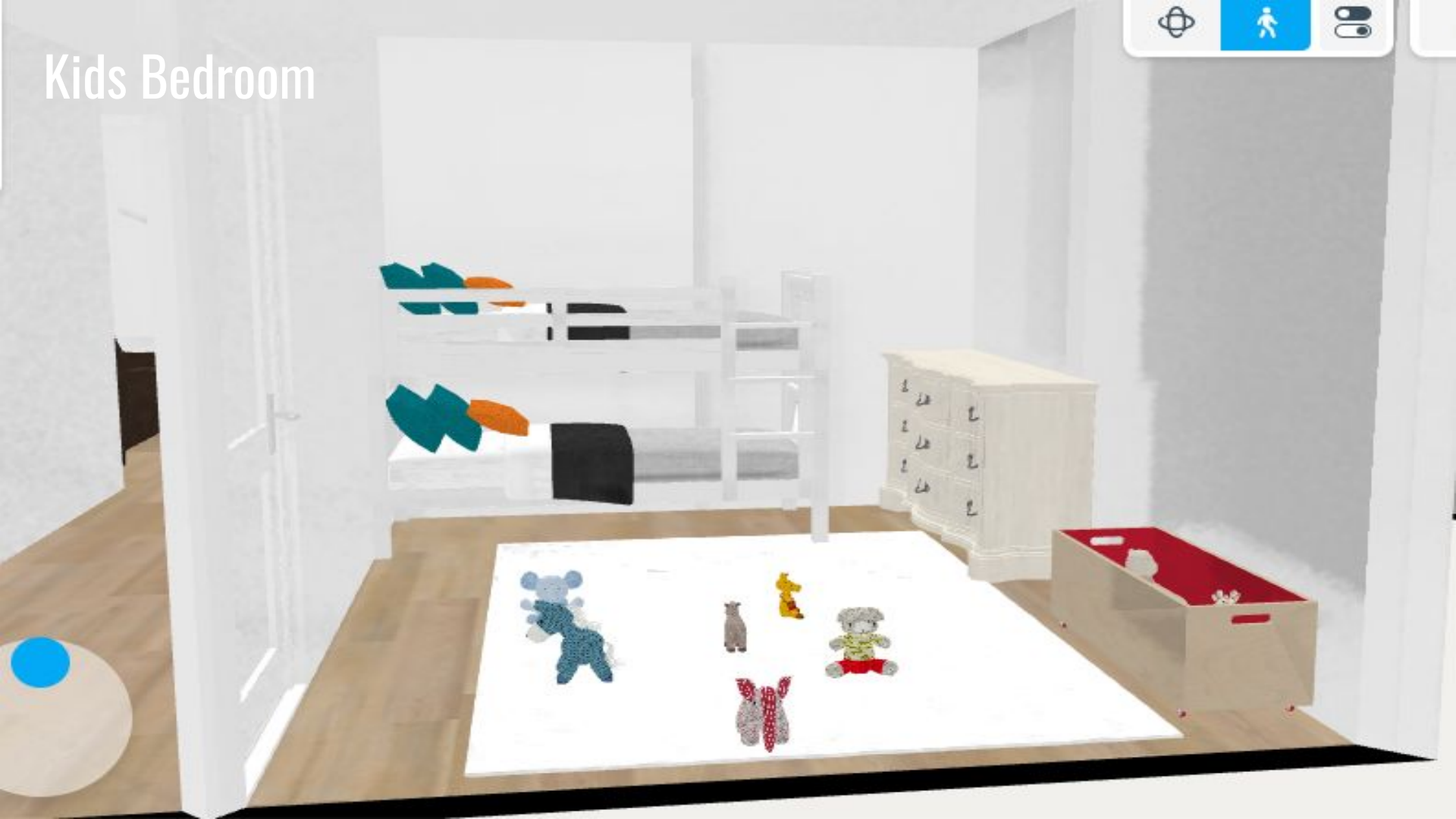
Master Bedroom



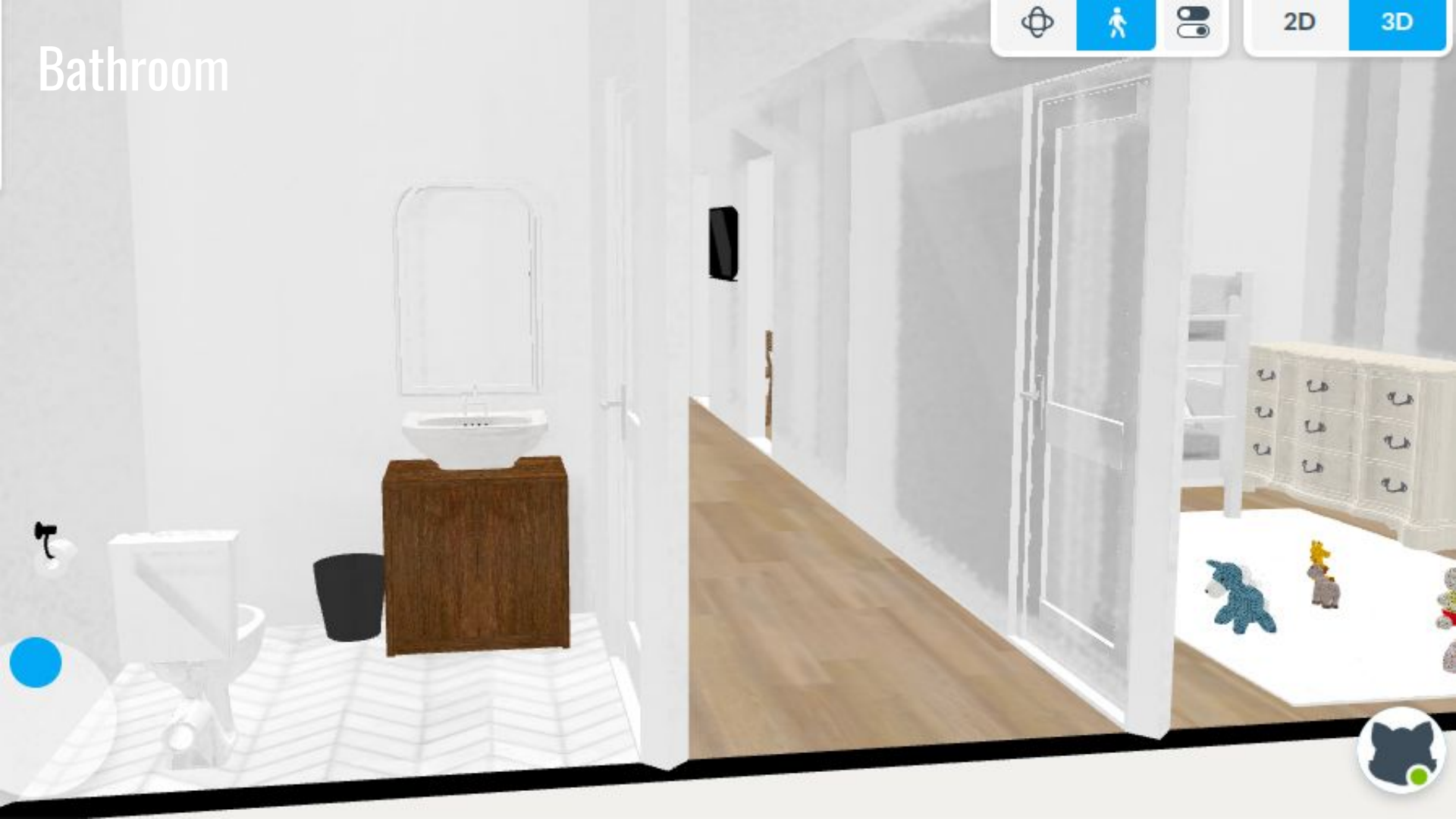
Master Bathroom



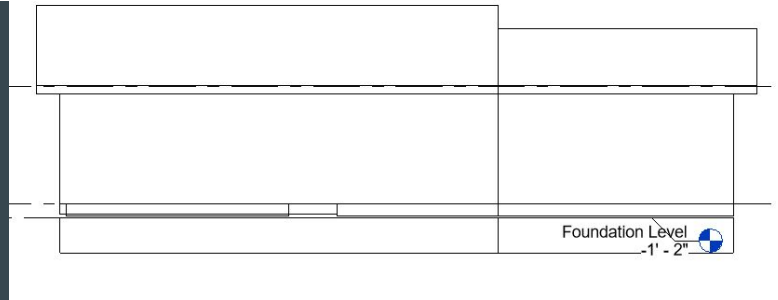
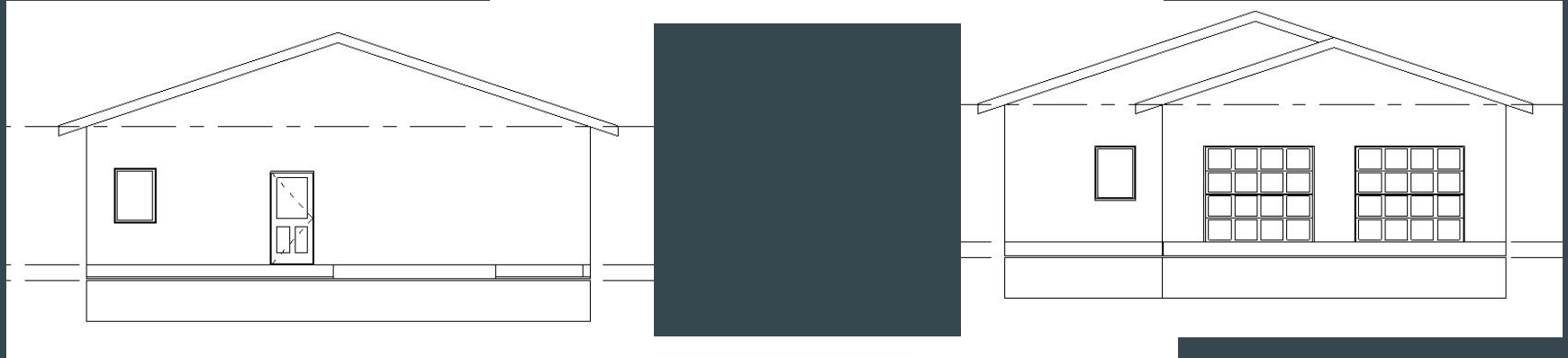
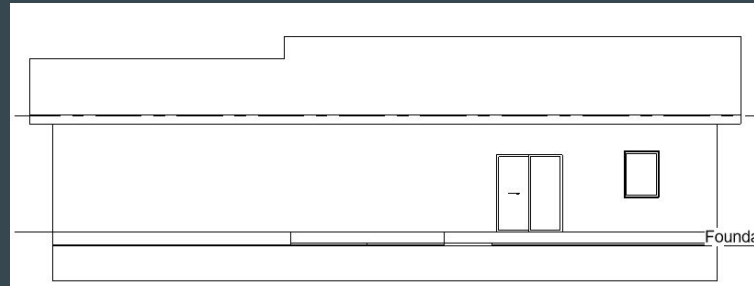
Kids Bedroom



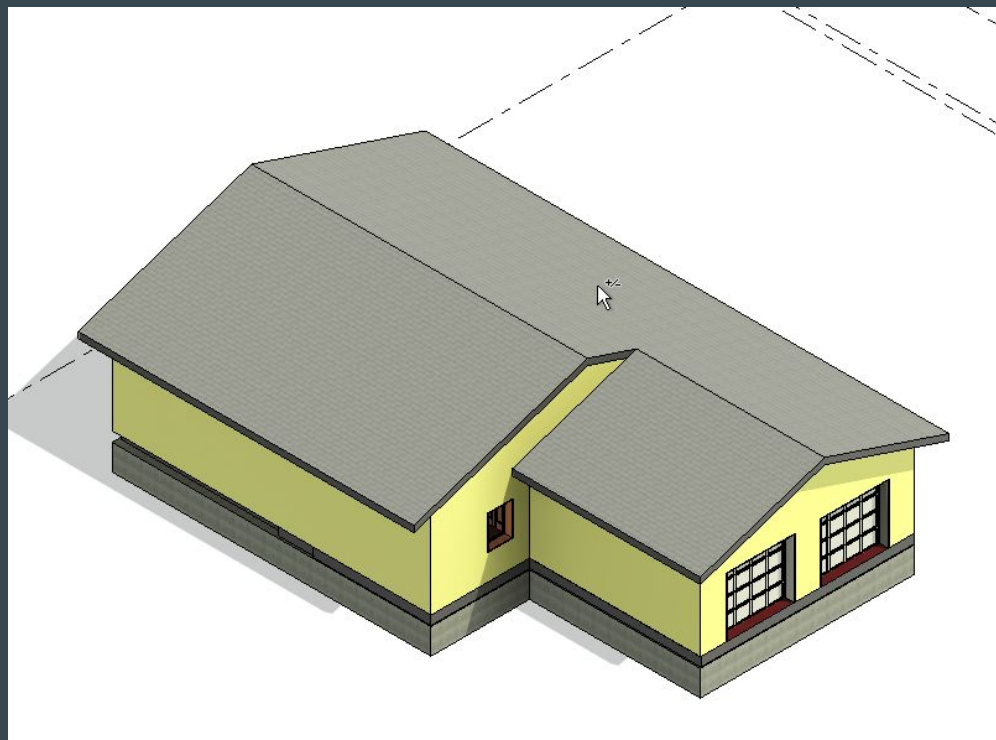
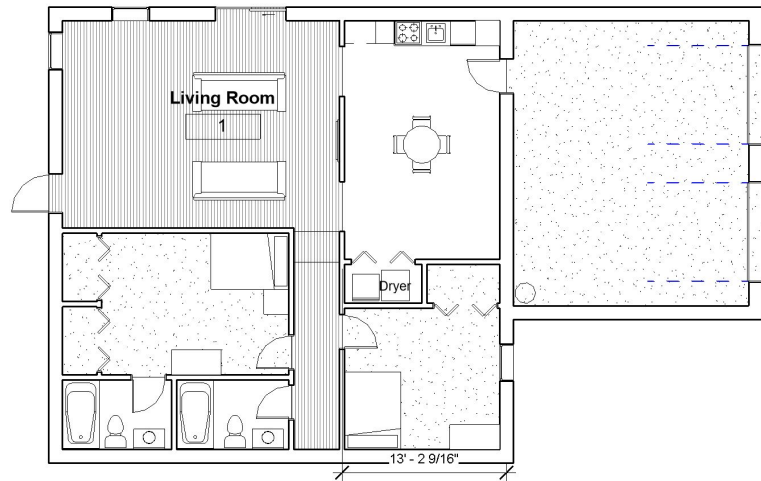
Bathroom



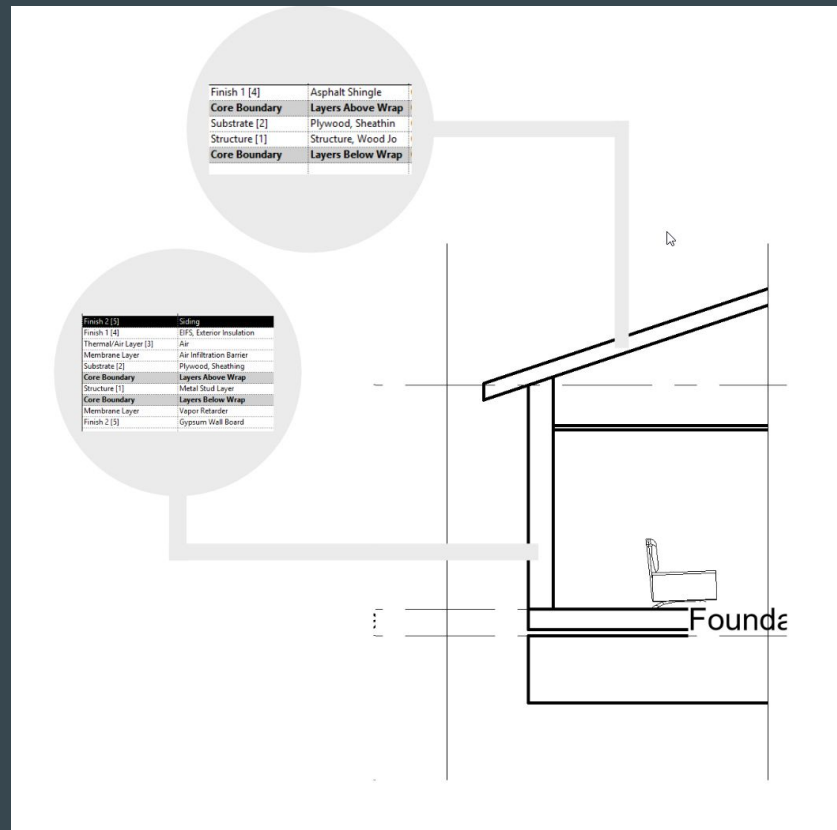
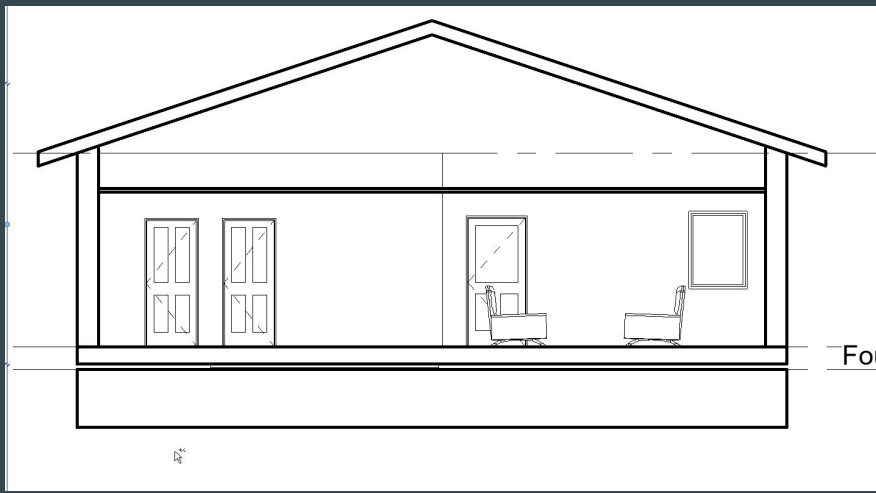
Revit



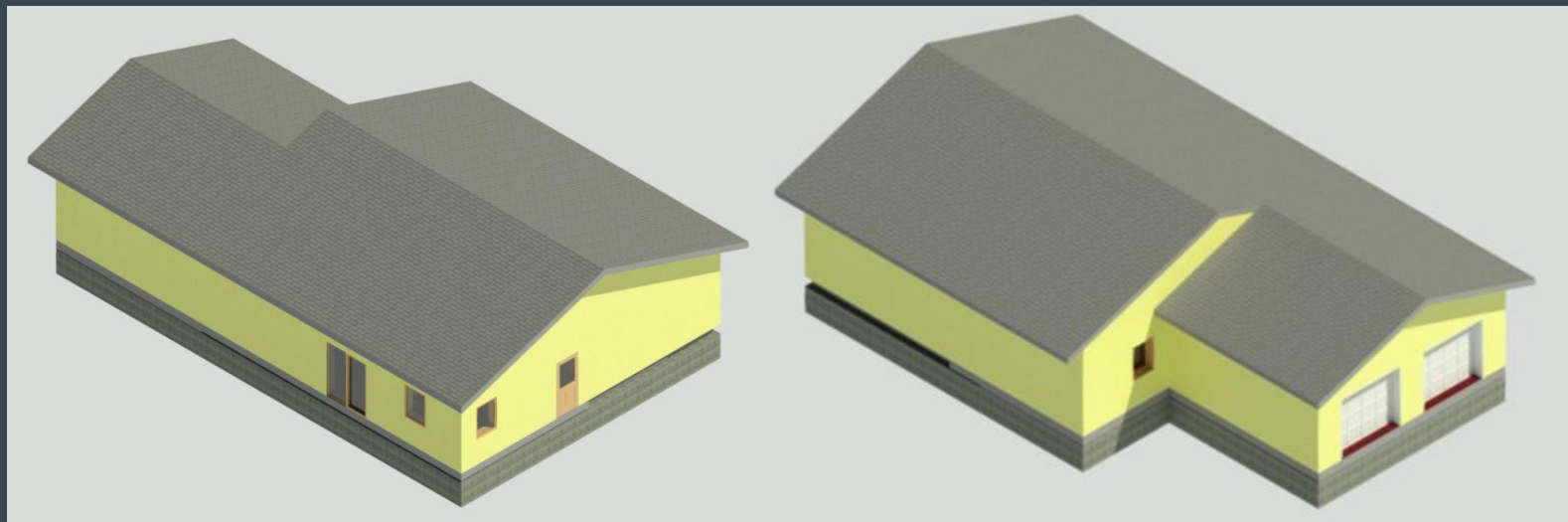
Elevations



Top and 3D

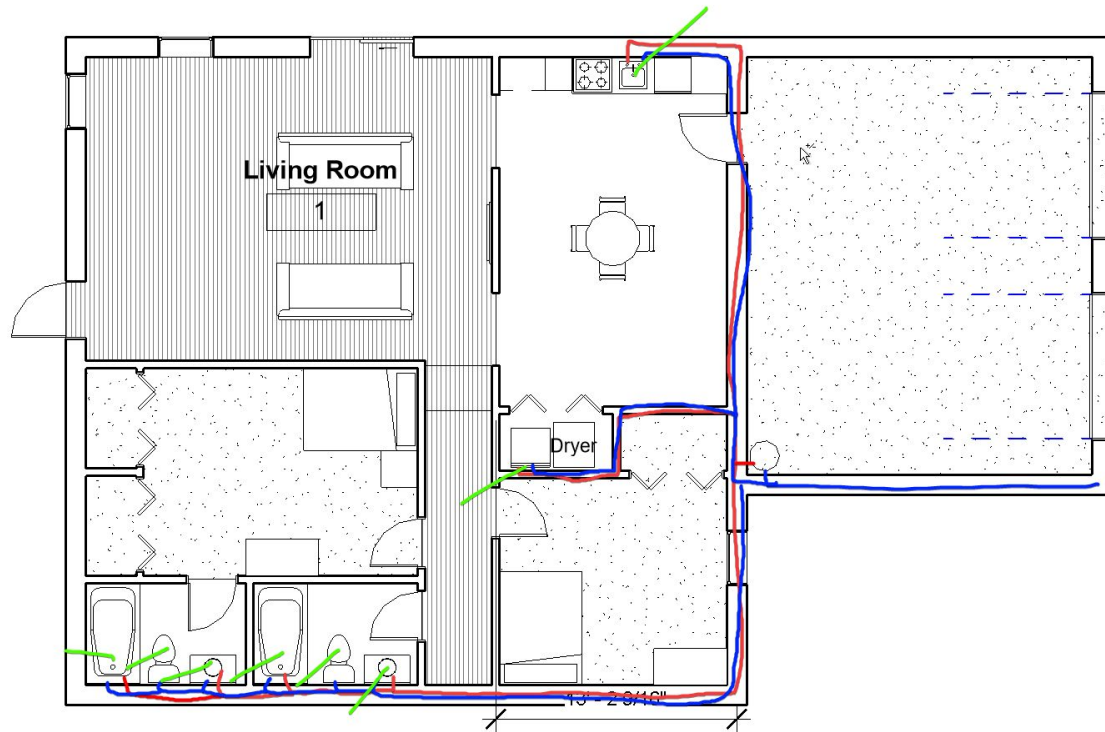


Section Views

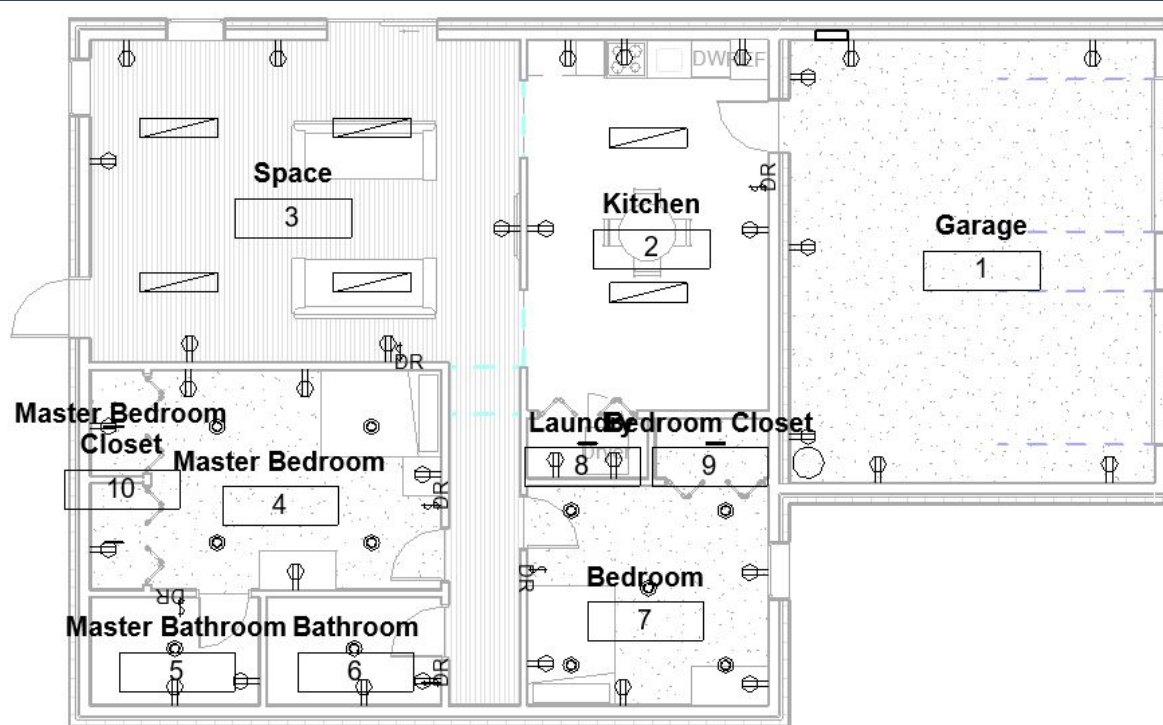


Renders

Plumbing



Electrical



R / W / L

Right - We built a house with most of the requirements. We made it a good-sized home. We tested many different designs and incorporated innovative elements to improve the energy efficiency of the home.

Wrong - We finished late. We failed to create a plumbing system in Revit due to the components properties.

Learned - We learned a lot more about Revit. How to do the electrical, render, and many more important parts of Revit. Also, what goes into planning and designing a house.